

Playwright.dev

API testing | Playwright Java

playwright.dev

Introduction

Playwright can be used to get access to the API of your application.

Sometimes you may want to send requests to the server directly from Java without loading a page and running js code in it. A few examples where it may come in handy:

Test your server API.

Prepare server side state before visiting the web application in a test.

Validate server side post-conditions after running some actions in the browser.

All of that could be achieved via methods.

Writing API Test

can send all kinds of HTTP(S) requests over network.

The following example demonstrates how to use Playwright to test issues creation via . The test suite will do the following:

Create a new repository before running tests.

Create a few issues and validate server state.

Delete the repository after running tests.

Configure

GitHub API requires authorization, so we'll configure the token once for all tests. While at it,

we'll also set the baseUrl to simplify the tests.

Write tests

Now that we initialized request object we can add a few tests that will create new issues in the repository.

Setup and teardown

These tests assume that repository exists. You probably want to create a new one before running tests and delete it afterwards. Use `@BeforeAll` and `@AfterAll` hooks for that.

Complete test example

Here is the complete example of an API test:

See [experimental](#) to automatically initialize Playwright objects and more.

Prepare server state via API calls

The following test creates a new issue via API and then navigates to the list of all issues in the project to check that it appears at the top of the list. The check is performed using .

Check the server state after running user actions

The following test creates a new issue via user interface in the browser and then checks via API if it was created:

Reuse authentication state

Web apps use cookie-based or token-based authentication, where authenticated state is stored as . Playwright provides method that can be used to retrieve storage state from an authenticated context and then create new

contexts with that state.

Storage state is interchangeable between `localStorage` and `sessionStorage`.
You can use it to log in via API calls and then create a new context with cookies already there.
The following code snippet retrieves state from an authenticated `context` and creates a new `context` with that state.

Select text only